

Homework 1

All-Reduce Operation

Introduction

This assignment explores collective communication in synchronous data-parallel deep learning. When training a model across multiple GPUs, each device computes the gradients of its local shard of the data. These gradients must then be **aggregated** across all devices ensuring that every GPU holds an identical global gradient before updating its local model parameters. This collective operation is known as **all-reduce**.

The two questions below ask you to compute the minimum communication time for all-reduce under two different GPU network topologies. Support each answer with a step-by-step solution.

Question 1. Decentralized All-Reduce

Consider a synchronous data-parallel setup with 4 GPUs, each holding a complete copy of the model and its data shard in GPU memory. The gradient size is 3.2 GB. The GPUs are connected via **point-to-point links** in a fully-connected topology, as illustrated in Figure 1. Each link supports a bandwidth of 32 GB/s in both directions simultaneously.

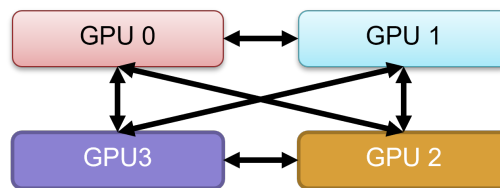


Figure 1: Fully-connected topology with 4 GPUs.

What is the minimum communication cost in milliseconds of the all-reduce operation in this fully-connected topology? Support your answer with a full derivation.

Step 1: Data partitioning. Each GPU is responsible for aggregating one-fourth of the global gradient, resulting in four equal partitions:

$$S_{\text{partition}} = \frac{3.2 \text{ GB}}{4} = 800 \text{ MB.}$$

Step 2: Parallel communication. Since each GPU has a point-to-point link to every other GPU, all partitions are transferred in parallel:

$$t_{\text{send}} = \frac{800 \text{ MB}}{32 \text{ GB/s}} = 25 \text{ ms.}$$

Step 3: Total communication cost. Once each GPU has locally aggregated its assigned partition, the reduced results are distributed back to all GPUs. In terms of communication cost, this receive phase is identical to the send phase:

$$t_{\text{total}} = t_{\text{send}} + t_{\text{receive}} = 25 \text{ ms} + 25 \text{ ms} = 50 \text{ ms.}$$

Question 2. Ring All-Reduce

Consider again 4 GPUs, each holding a full copy of the model and its data shard in GPU memory, with a gradient size of 3.2 GB. This time, however, the GPUs are arranged in a **ring topology**, as shown in Figure 2. Each link in the ring is unidirectional with a bandwidth of 32 GB/s.

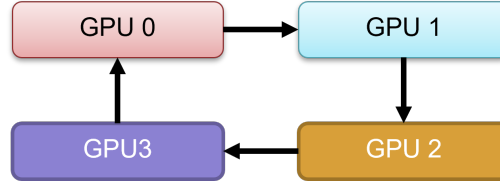


Figure 2: Ring topology with 4 GPUs.

What is the minimum communication cost in milliseconds of the all-reduce operation in this ring topology? Support your answer with a full derivation.

Step 1: Data partitioning. Each GPU is responsible for aggregating one-fourth of the global gradient, resulting in four equal partitions:

$$S_{\text{partition}} = \frac{3.2 \text{ GB}}{4} = 800 \text{ MB.}$$

Step 2: Sequential communication. The ring is unidirectional, meaning each partition must travel hop by hop to reach its destination GPU. The time to traverse a single hop is:

$$t_{\text{hop}} = \frac{800 \text{ MB}}{32 \text{ GB/s}} = 25 \text{ ms.}$$

Step 3: Bottleneck path. All GPUs begin sending simultaneously, so the bottleneck is determined by the longest path in the ring. The hop counts and latencies from GPU 0 are:

Path	Hops	Latency
GPU 0 → GPU 1	1	25 ms
GPU 0 → GPU 1 → GPU 2	2	50 ms
GPU 0 → GPU 1 → GPU 2 → GPU 3	3	75 ms

With $N = 4$ GPUs, the furthest destination requires $N - 1 = 3$ hops:

$$t_{\text{send}} = 3 \times 25 \text{ ms} = 75 \text{ ms.}$$

Step 4: Total communication cost. Once each GPU has locally aggregated its assigned partition, the reduced results must travel back around the ring. In terms of communication cost, this receive phase is identical to the send phase:

$$t_{\text{total}} = t_{\text{send}} + t_{\text{receive}} = 75 \text{ ms} + 75 \text{ ms} = 150 \text{ ms.}$$